



Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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SCIENCE

1113/02

Paper 2

April 2021

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

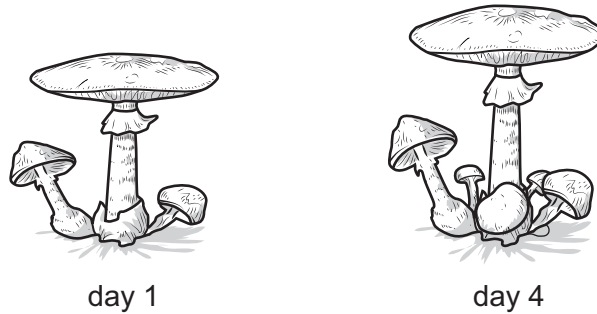
This document has **16** pages.



- 1 Yuri draws a picture of a fungus living near some trees.

After three days he draws another picture of the same fungus.

Look at these two pictures.



- (a) The pictures show evidence that this fungus is a living organism.

What is this evidence?

..... [1]

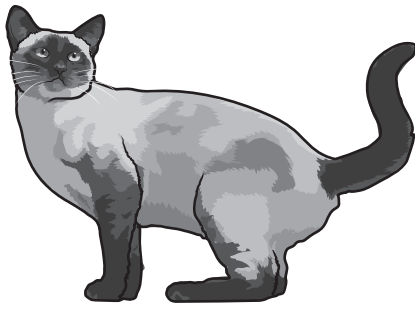
- (b) Yuri's teacher says that there are more characteristics of living organisms.

Write down three **other** characteristics of living organisms.

- 1
- 2
- 3

[3]

2 These varieties of cat belong to the same species.



Siamese cat



Manx cat

(a) Complete this definition of a species.

Animals that belong to the same species are able to together
to produce offspring.

[2]

(b) Siamese cats and Manx cats show variation within a species.

Describe **two** differences, shown in the diagram, between Siamese cats and Manx cats.

- 1
-
- 2
-

[2]

3 This question is about different types of chemical reaction.

(a) Complete the sentences.

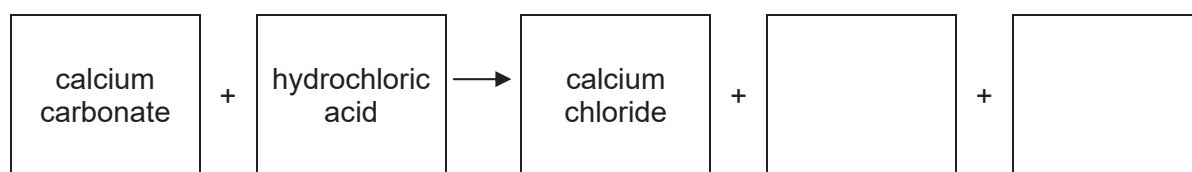
Choose from the list.

combustion displacement neutralisation respiration rusting

(i) Burning a fuel in air is called [1]

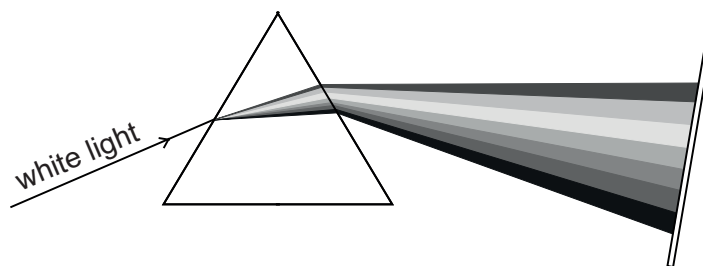
(ii) The reaction between an acid and an alkali is called [1]

(b) Complete the word equation for the reaction between calcium carbonate and hydrochloric acid.



[2]

- 4 Look at the diagram. It shows white light being split into different colours.



Complete the sentences.

Choose from the list.

dispersion	eight	filter	mirror
prism	reflection	refraction	screen
	seven	six	spectrum

When white light enters a it is split into
different colours.

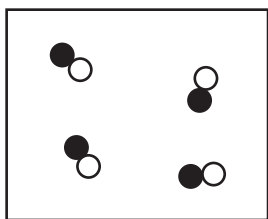
This process is known as

The different colours are known as the colours of the

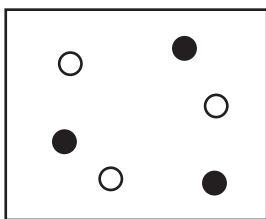
[4]

5 This question is about elements, compounds and mixtures.

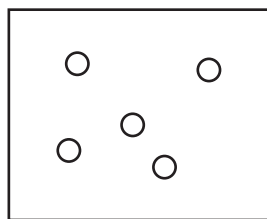
(a) Look at the boxes. They show the particles in some substances.



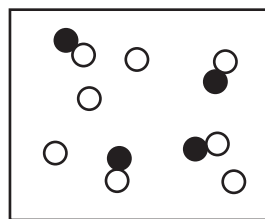
A



B



C



D

Which box shows a mixture of two elements?

Circle the correct answer.

A

B

C

D

[1]

(b) A compound is made of only sodium and chlorine.

What is the **name** of this compound?

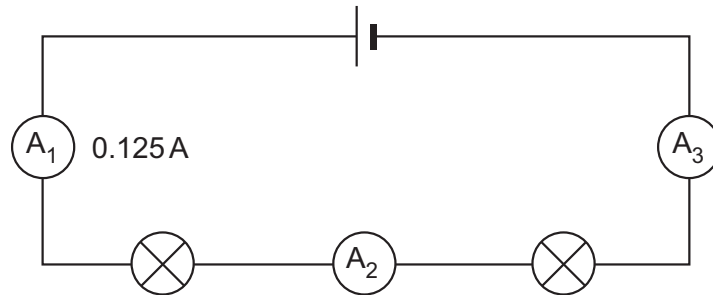
..... [1]

6 Mia measures the current in different electrical circuits.

(a) Write down the name of the piece of equipment used to measure current.

..... [1]

(b) Mia connects an electrical circuit with one cell and two lamps.



The reading on **A**₁ is 0.125 A.

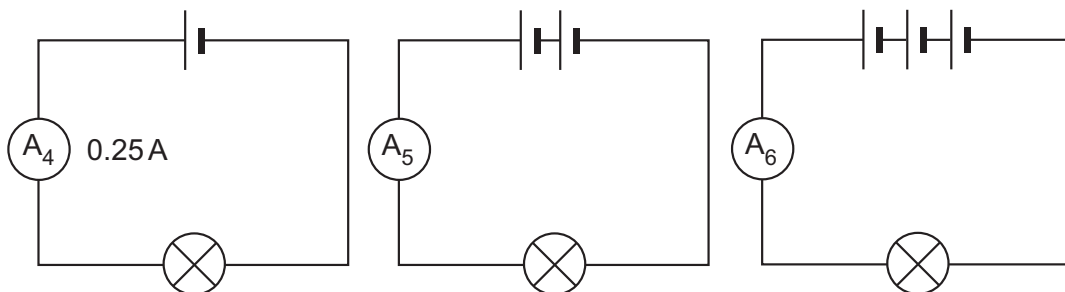
Write down the **two** missing readings.

The reading on **A**₂ is A.

The reading on **A**₃ is A.

[1]

(c) Mia connects **three** more circuits using the same size cells and lamps.



The reading on **A**₄ is 0.25 A.

Write down the **two** missing readings.

The reading on **A**₅ is A.

The reading on **A**₆ is A.

[2]

- 7 Chen measures the pulse rate of his friends before and after exercise.

All the friends do the same amount and level of exercise.

Here are his results.

friend	pulse rate before exercise in beats per minute	pulse rate after exercise in beats per minute	change in pulse rate in beats per minute
Carlos	74	104	
Mike	72	105	
Oliver	69	98	

- (a) Calculate the change in pulse rate for each friend.

Write your answers in the table.

[1]

- (b) Are Chen's results **reliable**?

Yes

☐

No

☐

Explain your answer.

..... [1]

- (c) Chen makes this statement.

'Mike is the fittest boy in my school.'

Write down **two** reasons why Chen's statement is **not** correct.

1

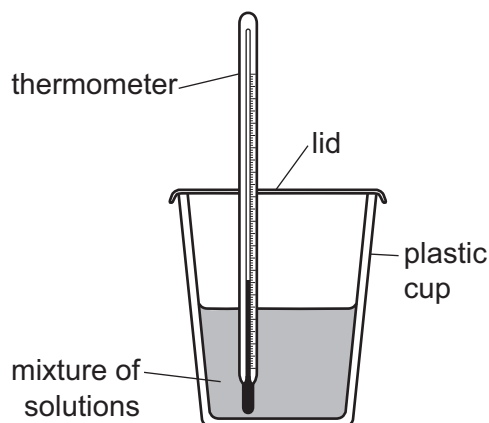
.....

2

.....

[2]

8 Look at the diagram. It shows solutions being mixed in a plastic cup.



Look at the table. It shows the results of five experiments.

experiment	temperature before mixing solutions in °C	temperature after mixing solutions in °C
A	15	26
B	15	10
C	15	15
D	15	32
E	15	27

(a) (i) Which experiment transfers the most thermal (heat) energy into the solutions?

..... [1]

(ii) Complete the sentence.

The reaction in experiment is **endothermic** because

..... [1]

(b) Suggest a reason for the result in experiment **C**.

..... [1]

- 9 Priya measures the distance a swimmer moves every five seconds.

She investigates two different swimming styles.

She writes her results in a table.

time in s	distance in m	
	butterfly style	backstroke style
0.0	0.0	0.0
5.0	8.5	9.0
10.0	16.5	18.0
15.0	25.0	34.5
20.0	33.0	35.0
25.0	42.0	46.0

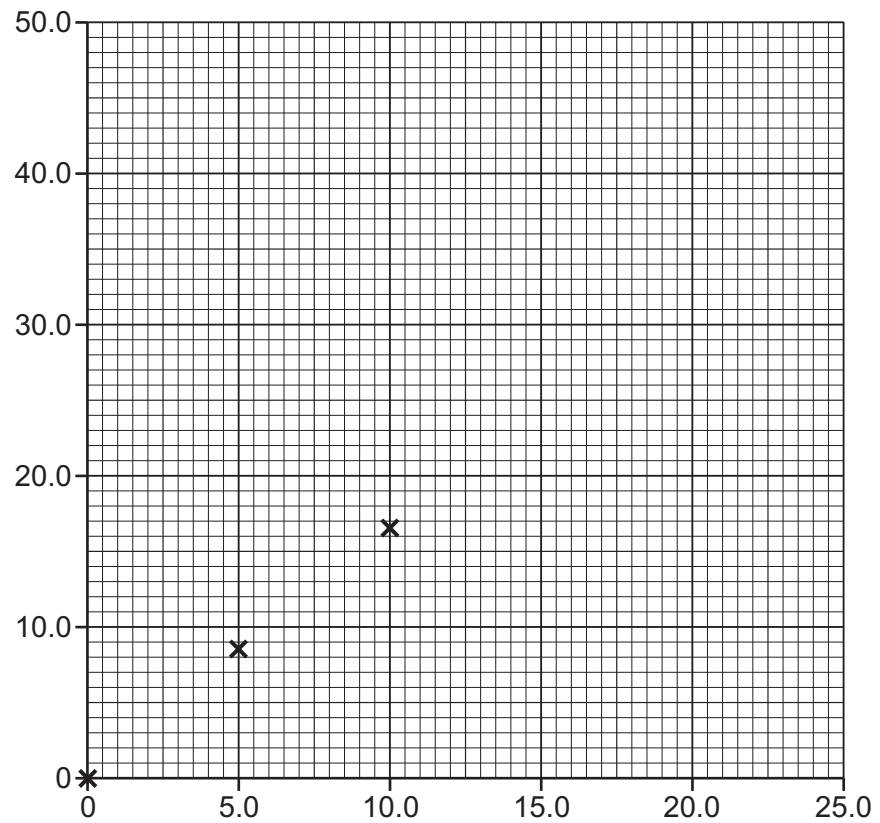
- (a) One of the results is **anomalous**.

Circle the **anomalous** result in the table.

[1]

(b) (i) Complete the distance/time graph for the **butterfly style**:

- label the x-axis and y-axis
- plot the other three points
- draw a line of best fit.



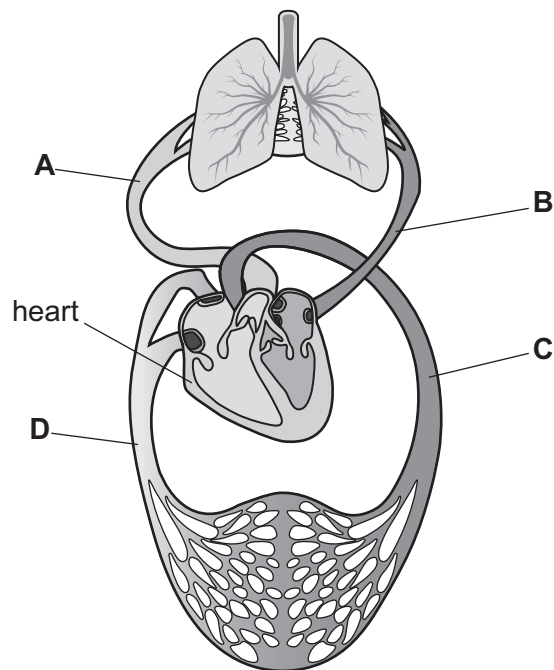
[3]

(ii) Predict the distance the swimmer moved in the first 12 seconds.

.....m [1]

10 The diagram shows the heart of a human.

The darker grey shading shows oxygenated blood.



Use the information in the diagram to match each **letter** with the correct **description of the blood vessel**.

letter	description of the blood vessel
A	artery taking deoxygenated blood to the lungs
B	artery taking oxygenated blood to the body
C	vein taking deoxygenated blood to the heart
D	vein taking oxygenated blood to the heart

[3]

11 Look at the chemical symbols for four metals.

Cu

Zn

Na

Ca

Answer the questions.

Each chemical symbol can be used once, more than once or not at all.

(a) Which of the four metals is the **most** reactive?

.....

[1]

(b) Which of the four metals does **not** react with water or dilute hydrochloric acid?

.....

[1]

(c) Which of the four metals reacts **slowly** with dilute hydrochloric acid?

.....

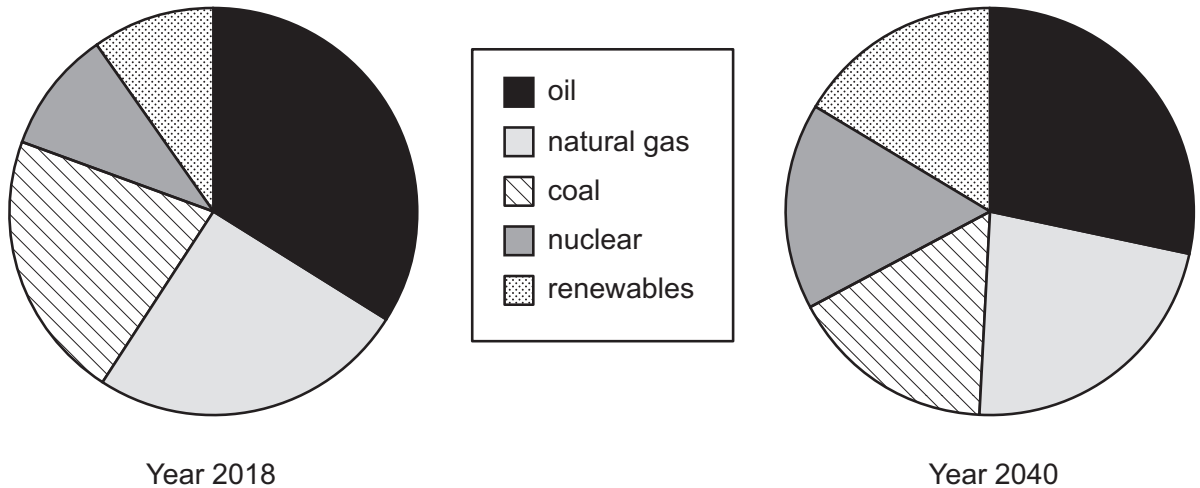
[1]

(d) Which of the four metals is **lowest** in the reactivity series?

.....

[1]

12 Safia draws two pie charts to show the energy resource use in her country.



The pie chart for the year 2040 shows the predicted energy resource use.

The predicted **natural gas** use decreases from 2018 to 2040.

The predicted **renewables** use increases from 2018 to 2040.

(a) Describe two **other** changes in predicted energy resource use from 2018 to 2040.

- 1

 2

 [2]

(b) (i) Name one renewable energy resource.

..... [1]

(ii) Suggest why the renewable energy resource use is predicted to **increase** from 2018 to 2040.

.....
 [1]

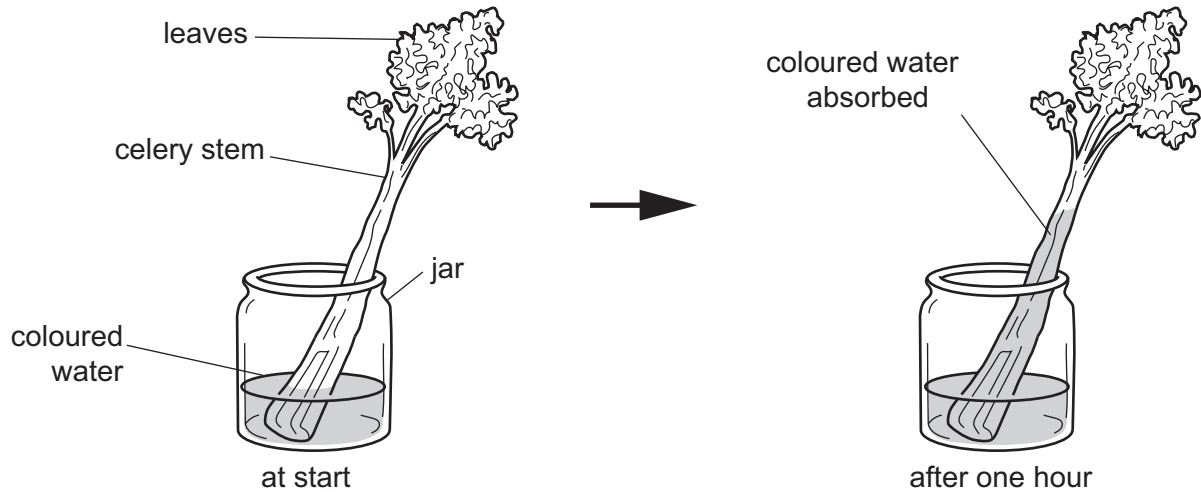
13 Angelique investigates how quickly plants absorb water.

She uses the stem and leaves of a plant called celery.

She puts a stem of celery in a jar of coloured water.

After one hour she looks at the celery to see how much coloured water it has absorbed.

This is shown in the diagram.



(a) Suggest how Angelique measures the amount of coloured water the celery has absorbed.

.....
 [1]

(b) Angelique also investigates the effect of temperature on the absorption of water.

State **two** variables that Angelique controls in this investigation.

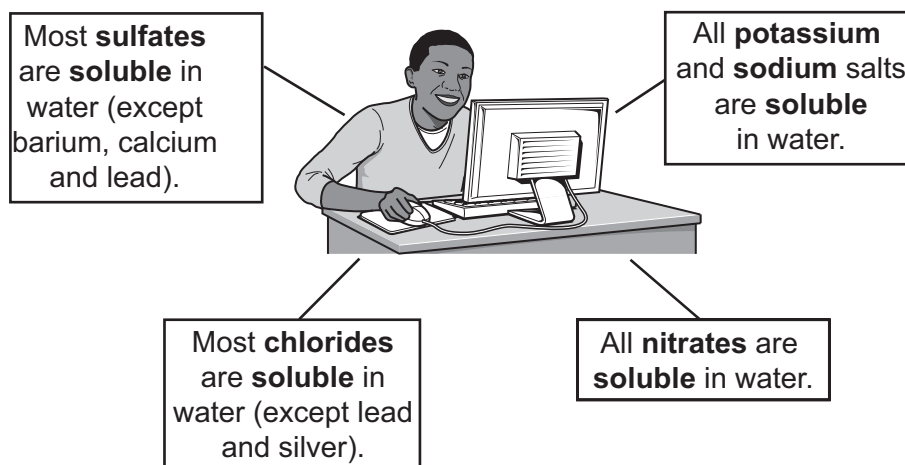
1

2

[2]

14 Carlos researches the solubility of different salts.

He finds this information on the internet.



Use the information to answer these questions.

(a) Write down the name of one **insoluble** sulfate.

..... [1]

(b) Look at the chemical formula of a salt.



Is the salt soluble?

Yes ☐ No ☐

Explain your answer.

..... [1]

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